

PROJECT 08: MOVIE AND CINEMA ROOM MANAGEMENT SYSTEM

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Project Objective

The objective of this project is to develop a database management system to efficiently manage movies, cinema rooms, screening schedules, customers, and ticket reservations. The system should improve booking experience, streamline cinema operations, and support report generation.

Database and Programming Languages

- **Database Management System:** MySQL
- **Programming Language:** Python

Detailed Project Requirements

System Analysis and Requirements

- Manage detailed information about movies, screening rooms, screening schedules, customers, and ticket bookings.

Main Functionalities

- Movie management (add/edit movie information, genres, duration).
- Cinema room management (track room names, seat layout, capacity).
- Screening schedule management (assign movies to rooms at specific times).
- Customer management (add/update customer information).
- Ticket booking and seat selection.
- Generate reports on ticket sales, screenings, and occupancy rates.

Database Design and Implementation

1. Data Model Design

- Design ER diagram for the cinema system.
- Convert the ER model into relational schema including PKs, FKs, and constraints.

2. Table Structures

- Movies (MovieID, MovieTitle, Genre, DurationMinutes)
- CinemaRooms (RoomID, RoomName, Capacity)
- Screenings (ScreeningID, MovieID, RoomID, ScreeningDate, ScreeningTime)
- Customers (CustomerID, CustomerName, PhoneNumber)
- Tickets (TicketID, CustomerID, ScreeningID, SeatNumber)

3. Sample Data

- Add 510 sample rows for each table.
- Create a database schema diagram using MySQL Workbench.

Advanced Database Objects

- **Indexes:** Speed up queries on movie titles, screening schedules.
- **Views:** Provide summaries of daily screenings or available seats.
- **Stored Procedures:** Automate ticket booking or check seat availability.
- **User Defined Functions:** Calculate occupancy rate or total revenue per screening.
- **Triggers:** Automatically update seat availability or notify overbooking.

Database Security and Administration

- Assign roles (admin, ticket clerk) and control access to sensitive data.
- Secure customer information and payment history (if included).
- Set up regular backups and recovery plans.
- Optimize screening lookup and ticket queries for performance.

Python Application Development

- **Database Connection:** Use `mysql-connector-python` or `SQLAlchemy`.
- **Booking System:** Write Python functions to handle seat booking and cancellation.
- **Reporting:** Generate screening and sales reports.
- **User Interface:** Build a simple command-line or GUI application for booking operations.

Deliverables

- A printed comprehensive report including database design, implementation details, and screenshots. Submit your report into LMS. First pages of your report must attach this PDF file.
- Link to publish Github source code and video presenting on Youtube
- SQL schema, sample data scripts, ER diagrams, and Python code.
- Demonstration of functionalities such as adding income, managing expenses, and generating reports.

Conclusion and Recommendations

- Evaluate the system's functionality and user experience.
- Suggest features for future enhancement such as online payments or QR-code-based tickets.

References

- Include all technical and academic resources used during the project.